Alma Models — Overnight validation report

Deeper tests of the frozen world models and of the integrity harness · 5 November 2026 · branch claude/gracious-franklin-f8ml84 @ edb02bd

Bottom line

Integrity (the harness): 12/12 planted fakes behaved as expected (caught, and the clean control passed); 27/27 tamper and no-training tests refused as expected; the exhaustive leak test checked 872 forecasts with corrupted futures and 45/45 rows behaved as expected, including 9 positive controls in which a deliberately leaky context WAS detected. **Physics/economics properties:** 4/4 held on real inputs. **Performance over the 16 months after training (Jul 2025 – Nov 2026):** pooled comparisons of each frozen Alma model against every competitor — significantly better / tie / significantly worse: oil 4/4/0 of 8; power 4/6/2 of 12; gas 11/1/0 of 12; crypto 12/1/1 of 14. Most of this period (VAL, TEST) had been scored before, so it is a re-look, not a fresh test; the fresh, sealed forward evidence is still accumulating (Part D).

Scorecard

Project	Leak tests (changed / forecasts)	Positive controls fired	Re-run = saved	Physics test	Pooled 16 months vs best external competitor	World model vs twin
gas	0 / 120	3 / 3	4 / 4	held	wm_gas_ens better than st_ols_wx (QS 4.2767 vs 6.1141)	tie
power	0 / 288	2 / 2	2 / 2	held	wm_power worse than chronos2 (QS 8.6021 vs 8.1517)	tie
crypto	0 / 272	2 / 2	2 / 2	held	wm_fin better than lgbm_dow (QS 0.1491 vs 0.1540)	twin better
oil	0 / 192	2 / 2	2 / 2	held	wm_sv tie with ewma94 (QS 0.00917 vs 0.00919)	tie

Leak tests: every Oct-Nov 2026 origin, four kinds of corrupted future (V3). Positive controls: a deliberately leaky context must be flagged. Re-run: saved forecasts equal a fresh run of the frozen model (deployment audit, section R). Property: the physics/economics check of V5. Pooled: Jul 2025 - Nov 2026, 95% block bootstrap on the quantile score.

How to read this report

Part	Question	Tests
A	Can the audit be fooled? Would it notice a leak, a fake number, a tampered model?	V1 mutation tests, V2 tamper + determinism, V3 exhaustive leak tests with positive controls, new re-run check (R)
B	Is the performance real and stable after training?	V4: every model, every month, Jul 2025 – Nov 2026; pooled block-bootstrap verdicts; world-model vs direct twin over 16 months
C	Are the forecasts sensible and honest about uncertainty?	V5: calibration, physics/economics property tests, placebo (shuffled outcomes)
D	What is still open?	sealed ex-ante forecasts, forward test, caveats

Part A — Can the integrity harness be fooled?

A1. Defects the validation found in our own harness — all fixed and re-verified

The point of testing the tester: these were found by tonight's tests, not assumed away. None changed a reported forecast or score; the snapshot was re-audited afterwards (57/57).

ID	Severity	Defect	Fix	Verified by
F-V1-1	high (integrity gap)	The audit recomputed every metric from the saved forecasts but did not re-run the frozen model for oil and power, so a fabricated forecast file with consistent metrics would have passed.	New audit section R (tools/deploy/audit_deploy.py): each frozen world model is re-run on sampled saved origins in every project and must match exactly.	V1 mutants oil_fabricated_consistent and gas_fabricated_consistent are caught only by R.
F-V2-1	medium (guard weaker than documented)	no_training() patched torch.optim.Optimizer.step only; concrete optimizers (SGD, Adam, ...) override step, so an optimizer step was not blocked (backward() was blocked, so no gradient could reach the weights, and parameter hashes were unchanged in every run).	no_training() now blocks optimizer construction, step() on every Optimizer subclass, Tensor.backward, torch.autograd.backward and torch.autograd.grad.	V2 re-run: T5 optimizer step refused (27/27).
F-V2-2	medium (check covered the wrong copy)	verify_checkpoints() checked the copies in winning_models/, but the oil, power, gas (wm_gas) and fin wrappers load their own copies under runs//search/ckpt/. A tampered loaded copy was still refused by the wrapper's own sha256 assert (T1), but not by the deployment-level check.	verify_checkpoints() also verifies every loaded copy under runs//search/ckpt/ against the same pins (oil 6, power 6, gas 2, fin 2 files). The loaded copies were byte-identical to the pins throughout, so no deployed result was affected.	V2 re-run: T2 refused for oil, power, gas and fin.
F-V2-3	test defect (not a harness defect)	The V2 test tree did not contain runs/fin/search/ckpt, so the fin tamper test was refused for the wrong reason (file not found).	The test tree now contains the files the fin wrapper loads, and the flip targets runs/fin/search/ckpt/frozen_wm_fin.pt.	V2 re-run: T1c fin loads, T1 fin refused with a sha256 mismatch.
F-V3-1	high (a leak test was blind)	The frozen gas wrappers memoise storage() by (target_week, origin) only: wm_gas keeps _cache, wm_gas_ens keeps _cache and a persistent _raw. The deployment audit's gas leak test (section E) and the first V3 gas run queried the same week twice on one model instance (clean, then corrupted), so the second answer came from the memo: '0 forecasts changed' was guaranteed and proved nothing. The positive control (a deliberately leaky context) also showed 0 changes, which exposed it. The 1 Nov audit's own gas leak test (audit/gas_ens_leak.py) builds a fresh model per call and is not affected. In normal use the memo is harmless (one week and origin always means one context).	Every test now clears the memo before each call (gas_nocache in tools/deploy/audit_deploy.py and tools/validate/*.py). tools/gas/wm_gas.py is sha256-frozen for the forward test and was not changed.	V3 re-run with every memo cleared: the positive controls fire 5/5 for wm_gas, wm_gas_direct and wm_gas_ens, and corrupting the future still changes 0 of 240 gas forecasts.

A2. Mutation tests (V1): plant one fake, the audit must fail

Planted fake	Must trip	Tripped	Result
no change	nothing (control)	nothing	PASS (no false alarm)
Report a better score than the model achieved (wm_sv October 1-day QS improved by 5%).	D	D	CAUGHT
Edit a sealed forecast after the fact without updating its seal.	G	D, G, R	CAUGHT
Sophisticated fake: change the forecasts, re-seal them AND recompute every reported QS consistently.	R	R	CAUGHT
Change a realised value (the EIA storage change of the first October week) to flatter the model.	C	C, D	CAUGHT

Planted fake	Must trip	Tripped	Result
Claim significance on 5 weeks of data.	F	F	CAUGHT
Replace the sealed wm_gas_ens October forecasts with better ones (closer to the truth), re-seal, and recompute MAE and QS consistently.	R	R	CAUGHT
Change realised prices (the labels) for one delivery day.	C	C, D	CAUGHT
Report that a network's parameters changed during deployment (i.e. it was trained).	B	B	CAUGHT
Shift realised log-RV labels of three days.	C	C, D	CAUGHT
Edit the wm_fin forecasts and re-seal them, but leave the metrics.	D, R	D, R	CAUGHT
Report a continual-learning update during deployment.	B	B	CAUGHT

Check kinds: B no-training, C truth re-derived from raw files, D metric recomputation, F sample-size rule, G sealing (sha256), R fresh re-run of the frozen model.

A3. Tamper and determinism tests (V2)

Test	Project	Expected	Result	Detail
T1c	oil	loads	as expected	control (untouched copy): LOADED
T1	oil	refused	as expected	1-byte flip in runs/commodity/search/ckpt/f_sv_none_s0.pt -> AssertionError: checkpoint /home/user/Alma-Models/runs/validation/tamper/tree/runs/commod
T2	oil	refused	as expected	verify_checkpoints after the flip -> refused: loaded checkpoint copy differs from its pin {'file': 'runs/commodity/search/ckpt/f_sv_none_s0.pt', 'sha2
T1c	power	loads	as expected	control (untouched copy): LOADED
T1	power	refused	as expected	1-byte flip in runs/power/search/ckpt/frz_phys_none_s0.pt -> AssertionError: sha256 mismatch for runs/power/search/ckpt/frz_phys_none_s0.pt: 9b3d4d16d
T2	power	refused	as expected	verify_checkpoints after the flip -> refused: loaded checkpoint copy differs from its pin {'file': 'runs/power/search/ckpt/frz_phys_none_s0.pt', 'sha2
T1c	gas	loads	as expected	control (untouched copy): LOADED
T1	gas	refused	as expected	1-byte flip in winning_models/gas_wm_gas_ens/checkpoints/x_wm_s3456.pt -> AssertionError: sha256 mismatch for winning_models/gas_wm_gas_ens/checkpoint
T2	gas	refused	as expected	verify_checkpoints after the flip -> refused: checkpoint pin mismatch {'file': 'winning_models/gas_wm_gas_ens/checkpoints/x_wm_s3456.pt', 'sha256': '4
T1c	fin	loads	as expected	control (untouched copy): LOADED
T1	fin	refused	as expected	1-byte flip in runs/fin/search/ckpt/frozen_wm_fin.pt -> RuntimeError: wm_fin: checkpoint runs/fin/search/ckpt/frozen_wm_fin.pt sha256 2664ab1ebfc14141
T2	fin	refused	as expected	verify_checkpoints after the flip -> refused: loaded checkpoint copy differs from its pin {'file': 'runs/fin/search/ckpt/frozen_wm_fin.pt', 'sha256':
T3	power	refused	as expected	edit tools/power/wm_power.py -> refused: frozen code changed: tools/power/wm_power.py
T3	gas	refused	as expected	edit tools/gas/wm_gas.py -> refused: frozen code changed: tools/gas/wm_gas.py
T3	fin	refused	as expected	edit tools/fin/wm_fin.py -> refused: frozen code changed: tools/fin/wm_fin.py
T4	fin	refused	as expected	pinned file whose bytes do not match -> exit 1: runs/validation/tamper/manifest_with_wrong_pin.json: {'diff-cached': 1}; failures: 1
T4c	fin	accepted	as expected	control, correct pin -> exit 0
T5	all	refused	as expected	optimizer step under no_training() -> common.TrainingAttempted: training is disabled in deployment (tools/deploy/common.no_training)
T5	all	refused	as expected	backward() under no_training() -> common.TrainingAttempted: training is disabled in deployment (tools/deploy/common.no_training)
D1	oil	bit-identical	as expected	two runs, 1 thread: sha 6aa685ac05483fa3 vs 6aa685ac05483fa3

Test	Project	Expected	Result	Detail
D2	oil	report	as expected	1 vs 4 threads: bit-identical
D1	power	bit-identical	as expected	two runs, 1 thread: sha d72f0ed7a2341f3e vs d72f0ed7a2341f3e
D2	power	report	as expected	1 vs 4 threads: bit-identical
D1	gas	bit-identical	as expected	two runs, 1 thread: sha df7be700ca29937f vs df7be700ca29937f
D2	gas	report	as expected	1 vs 4 threads: bit-identical
D1	fin	bit-identical	as expected	two runs, 1 thread: sha bd2268ce92173419 vs bd2268ce92173419
D2	fin	report	as expected	1 vs 4 threads: bit-identical

T1 one-byte flip in a checkpoint → the wrapper refuses; T2 → the deployment refuses; T3 one-character edit of frozen model code → refused; T4 a raw data file that no longer matches its pin → refused; T5 optimizer step / backward() under the no-training lock → refused. D1 two runs → bit-identical forecasts; D2 1 vs 4 CPU threads (reported only).

A4. Exhaustive leak tests with positive controls (V3)

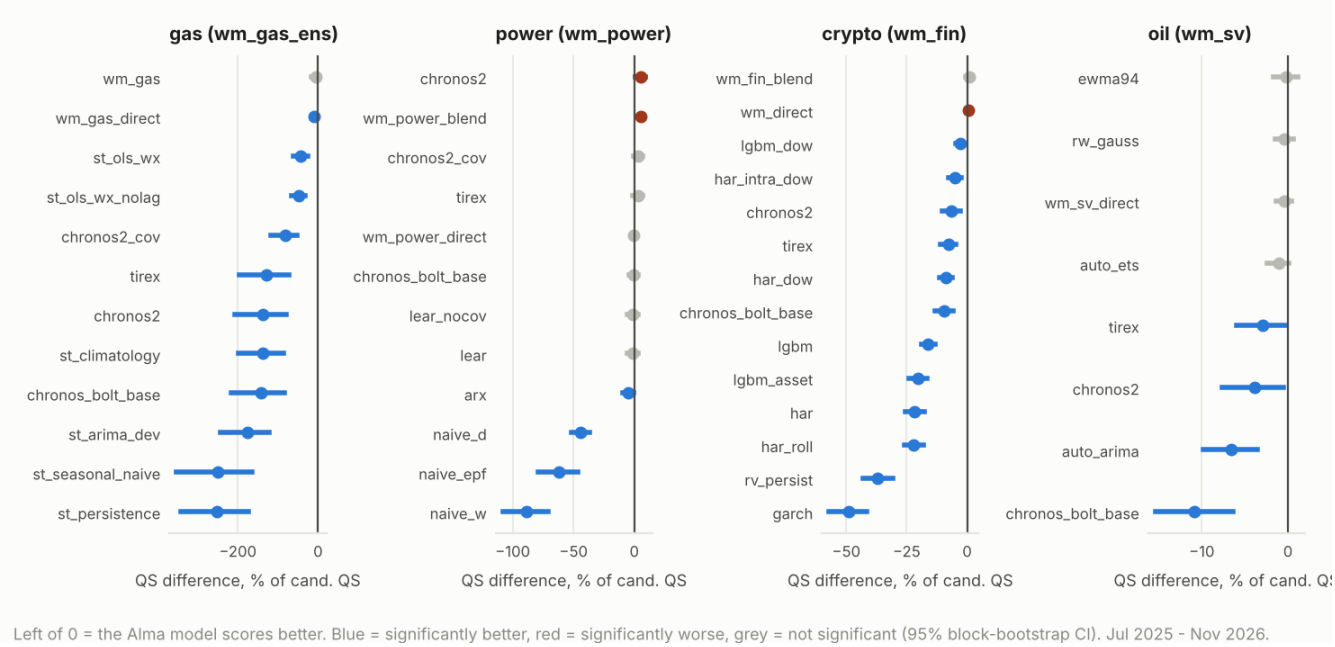
Every October–November 2026 origin (not a sample), every world model and twin, four kinds of corrupted future (random noise, all missing, ×1000, the past copied into the future). A forecast must not change. The positive controls use a deliberately leaky context (an off-by-one that lets the target in); the same test must flag those, proving it is not blind.

Project	Model	Corruption	Forecasts	Changed	Kind	Result
oil	wm_sv	noise	24	0	leak test	as expected
oil	wm_sv	nan	24	0	leak test	as expected
oil	wm_sv	extreme	24	0	leak test	as expected
oil	wm_sv	shift	24	0	leak test	as expected
oil	wm_sv	off-by-one	23	23	positive control	as expected
oil	wm_sv_direct	noise	24	0	leak test	as expected
oil	wm_sv_direct	nan	24	0	leak test	as expected
oil	wm_sv_direct	extreme	24	0	leak test	as expected
oil	wm_sv_direct	shift	24	0	leak test	as expected
oil	wm_sv_direct	off-by-one	23	23	positive control	as expected
gas	wm_gas	noise	10	0	leak test	as expected
gas	wm_gas	nan	10	0	leak test	as expected
gas	wm_gas	extreme	10	0	leak test	as expected
gas	wm_gas	shift	10	0	leak test	as expected
gas	wm_gas	observed-wx	5	5	positive control	as expected
gas	wm_gas_direct	noise	10	0	leak test	as expected
gas	wm_gas_direct	nan	10	0	leak test	as expected
gas	wm_gas_direct	extreme	10	0	leak test	as expected
gas	wm_gas_direct	shift	10	0	leak test	as expected
gas	wm_gas_direct	observed-wx	5	5	positive control	as expected
gas	wm_gas_ens	noise	10	0	leak test	as expected
gas	wm_gas_ens	nan	10	0	leak test	as expected
gas	wm_gas_ens	extreme	10	0	leak test	as expected
gas	wm_gas_ens	shift	10	0	leak test	as expected
gas	wm_gas_ens	observed-wx	5	5	positive control	as expected
power	wm_power	noise	36	0	leak test	as expected
power	wm_power	nan	36	0	leak test	as expected
power	wm_power	extreme	36	0	leak test	as expected
power	wm_power	shift	36	0	leak test	as expected
power	wm_power	includes-D	35	35	positive control	as expected
power	wm_power_direct	noise	36	0	leak test	as expected
power	wm_power_direct	nan	36	0	leak test	as expected
power	wm_power_direct	extreme	36	0	leak test	as expected

Project	Model	Corruption	Forecasts	Changed	Kind	Result
power	wm_power_direct	shift	36	0	leak test	as expected
power	wm_power_direct	includes-D	35	35	positive control	as expected
fin	wm_fin	noise	34	0	leak test	as expected
fin	wm_direct	noise	34	0	leak test	as expected
fin	wm_fin	nan	34	0	leak test	as expected
fin	wm_direct	nan	34	0	leak test	as expected
fin	wm_fin	extreme	34	0	leak test	as expected
fin	wm_direct	extreme	34	0	leak test	as expected
fin	wm_fin	shift	34	0	leak test	as expected
fin	wm_direct	shift	34	0	leak test	as expected
fin	wm_fin	includes-D	11	10	positive control	as expected
fin	wm_direct	includes-D	11	10	positive control	as expected

Part B — Is the performance real and stable after training?

Every frozen model and every pre-registered competitor was re-run on every forecast origin from 2025-07-01 (the day after the training data end) to the November 2026 data cutoff — about 16 months — with exactly the deployment harness (no retraining). Windows: each calendar month, VAL (2025-07..10), TEST (2025-11..2026-10), FORWARD (2026-11) and ALL. Pooled verdicts use a 95% block bootstrap (7-day blocks for daily data, overlap-aware) and are issued only with ≥ 10 independent units. **Caveat:** VAL and TEST were scored by the original project, so this is a re-look at known data, not a fresh test; it answers 'is the advantage stable month to month?', not 'was it pre-registered?'.



Left of 0 = the Alma model scores better. Blue = significantly better, red = significantly worse, grey = not significant (95% block-bootstrap CI). Jul 2025 - Nov 2026.

Figure B1. Pooled Jul 2025 – Nov 2026 quantile score: each Alma model minus each competitor.

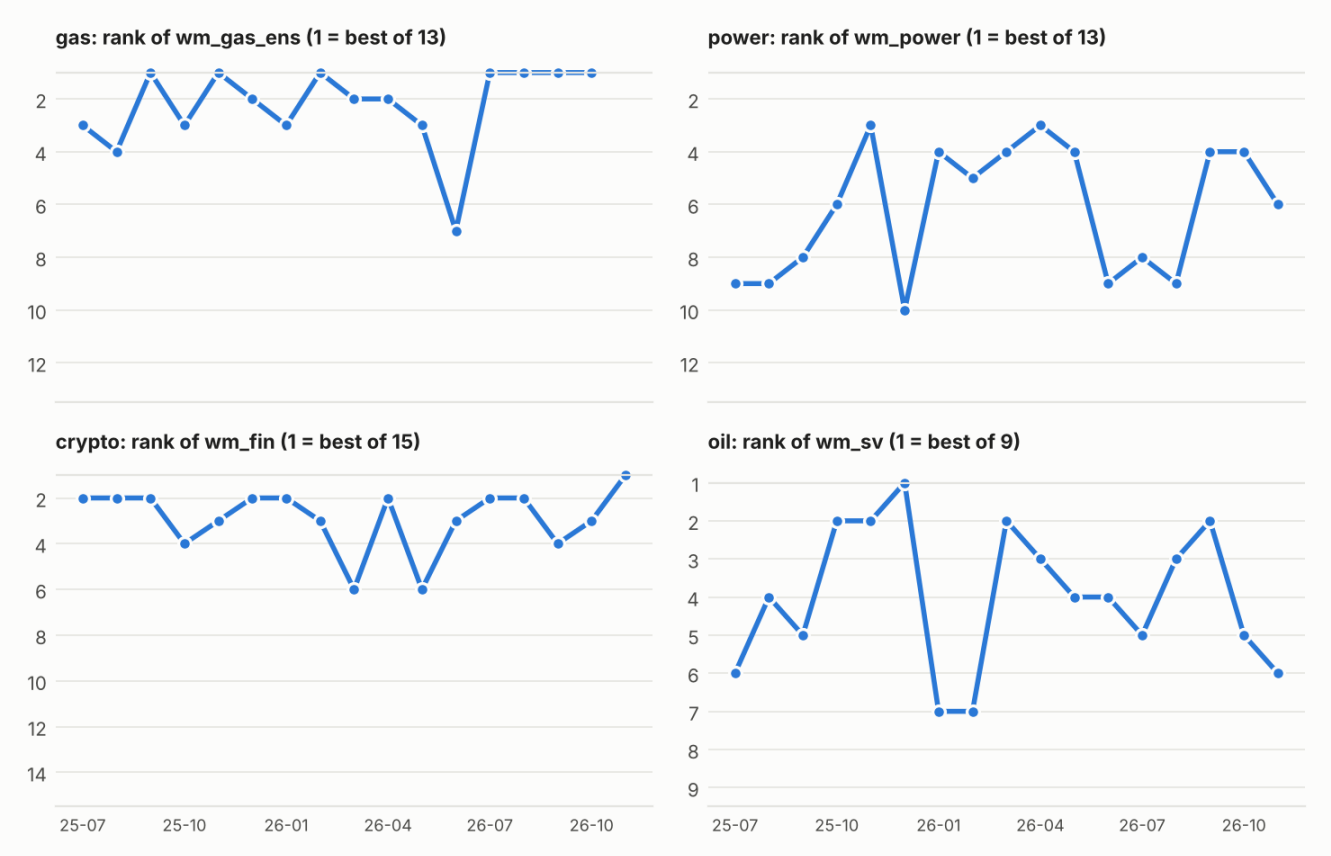


Figure B2. Month-by-month rank of each frozen Alma model among all models (1 = best).

B — gas: pooled Jul 2025 – Nov 2026 (n = 70 weeks)

Model	QS	diff vs wm_gas_ens	95% CI	n_eff	verdict
wm_gas_ens	4.28	–	–	–	(candidate)
wm_gas	4.54	-0.27	[-0.69, 0.12]	70	tie
wm_gas_direct	4.65	-0.37	[-0.68, -0.06]	70	wm_gas_ens better
st_ols_wx	6.11	-1.84	[-2.65, -1.09]	70	wm_gas_ens better
st_ols_wx_nolag	6.33	-2.05	[-2.78, -1.36]	70	wm_gas_ens better
chronos2_cov	7.80	-3.52	[-5.00, -2.21]	70	wm_gas_ens better
tirex	9.78	-5.50	[-8.40, -3.10]	70	wm_gas_ens better
chronos2	10.15	-5.88	[-8.89, -3.43]	70	wm_gas_ens better
st_climatology	10.19	-5.91	[-8.46, -3.72]	70	wm_gas_ens better
chronos_bolt_base	10.39	-6.11	[-9.25, -3.58]	70	wm_gas_ens better
st_arima_dev	11.80	-7.53	[-10.41, -5.20]	70	wm_gas_ens better
st_seasonal_naive	14.94	-10.66	[-15.10, -7.04]	70	wm_gas_ens better
st_persistence	15.06	-10.78	[-14.64, -7.47]	70	wm_gas_ens better

B — power: pooled Jul 2025 – Nov 2026 (n = 493 days)

Model	QS	diff vs wm_power	95% CI	n_eff	verdict
chronos2	8.15	0.45	[0.10, 0.78]	70	wm_power worse
wm_power_blend	8.17	0.43	[0.23, 0.63]	70	wm_power worse
chronos2_cov	8.34	0.26	[-0.05, 0.58]	70	tie
tirex	8.36	0.24	[-0.10, 0.56]	70	tie
wm_power	8.60	–	–	–	(candidate)
wm_power_direct	8.63	-0.03	[-0.09, 0.03]	70	tie
chronos_bolt_base	8.64	-0.04	[-0.36, 0.28]	70	tie
lear_nocov	8.68	-0.08	[-0.49, 0.29]	70	tie
lear	8.69	-0.09	[-0.50, 0.29]	70	tie
arx	9.01	-0.41	[-0.78, -0.05]	70	wm_power better
naive_d	12.40	-3.80	[-4.45, -3.16]	70	wm_power better
naive_epf	13.93	-5.33	[-6.78, -4.02]	70	wm_power better
naive_w	16.20	-7.60	[-9.31, -6.09]	70	wm_power better

B — crypto: pooled Jul 2025 – Nov 2026 (n = 491 days)

Model	QS	diff vs wm_fin	95% CI	n_eff	verdict
wm_fin_blend	0.1480	0.0011	[-0.0006, 0.0028]	69	tie
wm_direct	0.1483	0.0008	[0.0000, 0.0017]	70	wm_fin worse
wm_fin	0.1491	–	–	–	(candidate)
lgbm_dow	0.1540	-0.0042	[-0.0071, -0.0013]	70	wm_fin better
har_intra_dow	0.1568	-0.0077	[-0.0116, -0.0038]	69	wm_fin better
chronos2	0.1593	-0.0096	[-0.0152, -0.0045]	70	wm_fin better
tirex	0.1612	-0.0115	[-0.0166, -0.0067]	70	wm_fin better
har_dow	0.1621	-0.0130	[-0.0169, -0.0091]	70	wm_fin better
chronos_bolt_base	0.1638	-0.0141	[-0.0198, -0.0087]	70	wm_fin better
lgbm	0.1739	-0.0240	[-0.0283, -0.0195]	70	wm_fin better
lgbm_asset	0.1801	-0.0303	[-0.0357, -0.0248]	70	wm_fin better
har	0.1813	-0.0322	[-0.0378, -0.0263]	70	wm_fin better
har_roll	0.1820	-0.0330	[-0.0386, -0.0268]	70	wm_fin better
rv_persist	0.2041	-0.0551	[-0.0641, -0.0460]	70	wm_fin better
garch	0.2224	-0.0728	[-0.0852, -0.0620]	70	wm_fin better

B — oil: pooled Jul 2025 – Nov 2026 (n = 337 days)

Model	QS	diff vs wm_sv	95% CI	n_eff	verdict
wm_sv	0.00917	–	–	–	(candidate)
ewma94	0.00919	-0.00002	[-0.00015, 0.00011]	337	tie
rw_gauss	0.00920	-0.00003	[-0.00014, 0.00006]	337	tie
wm_sv_direct	0.00921	-0.00004	[-0.00012, 0.00004]	337	tie
auto_ets	0.00926	-0.00010	[-0.00022, 0.00001]	337	tie
tirex	0.00943	-0.00027	[-0.00055, -0.00002]	337	wm_sv better
chronos2	0.00952	-0.00036	[-0.00070, -0.00005]	337	wm_sv better
auto_arima	0.00977	-0.00061	[-0.00090, -0.00033]	337	wm_sv better
chronos_bolt_base	0.01016	-0.00100	[-0.00141, -0.00058]	337	wm_sv better

B5. By split

Project	Split	n	Alma QS	best competitor	rank
gas	VAL	18	4.66	tirex 5.67	2/13
gas	TEST	52	4.14	st_ols_wx 6.14	1/13
gas	ALL	70	4.28	st_ols_wx 6.11	1/13
power	VAL	123	7.32	chronos2 6.73	9/13
power	TEST	365	9.02	chronos2 8.67	5/13
power	FORWARD	5	9.42	chronos2 5.59	6/13
power	ALL	493	8.60	chronos2 8.15	5/13
crypto	VAL	123	0.1543	lgbm_dow 0.1590	3/15
crypto	TEST	365	0.1476	lgbm_dow 0.1526	3/15
crypto	FORWARD	3	0.1104	chronos2 0.1194	1/13
crypto	ALL	491	0.1491	lgbm_dow 0.1540	3/15
oil	VAL	86	0.00520	rw_gauss 0.00520	3/9
oil	TEST	249	0.01047	ewma94 0.01050	1/9
oil	FORWARD	2	0.01766	chronos2 0.01560	6/9
oil	ALL	337	0.00917	ewma94 0.00919	1/9

B6. The world-model test over 16 months: rollout vs matched direct twin

Project	World model QS	Direct twin QS	WM − twin	95% CI	verdict
oil	0.00917	0.00921	-0.00004	[-0.00012, 0.00004]	tie
power	8.6021	8.6309	-0.0288	[-0.0877, 0.0297]	tie
gas	4.5449	4.6497	-0.1048	[-0.4592, 0.2743]	tie
crypto	0.1491	0.1483	0.0008	[0.0000, 0.0017]	twin better
power (P2 hourly direction, higher = better)	0.7416	0.7334	0.0082	[0.0028, 0.0139]	rollout better

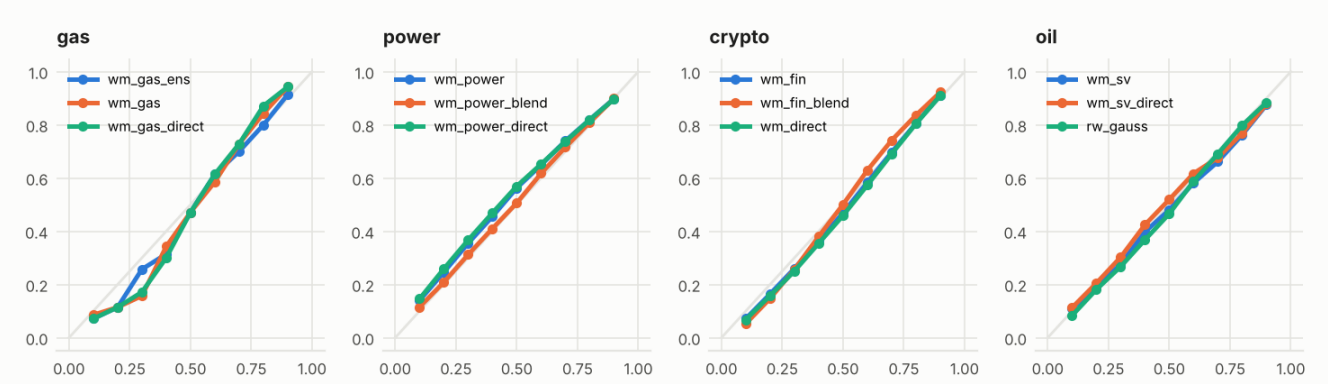
Quantile score rows: lower is better. Across the four projects the carried state is a tie in three and slightly worse in crypto; on the power project's pre-registered twin metric (hourly direction) the rollout is significantly better over the 16 months. Several metrics are shown, so the single positive result should be read with that multiplicity in mind.

Part C — Are the forecasts sensible and honest about uncertainty?

C1. Physics / economics property tests (frozen models, real October 2026 inputs)

Project	Property	Cases	Violations	Measured effect
power	less wind never lowers any price quantile (merit order)	48360	0	mean median price by wind change {"-0.5": 185.16, "-0.3": 181.32, "-0.1": 177.13, "0.0": 174.96, "0.1": 172.78, "0.3": 1
gas	colder forecast (+HDD) never raises predicted injection	30	0	mean change in the weekly median for +20 HDD/day on forecast days: -34.2 Bcf
oil	twice-as-volatile recent returns widen the 80% interval	125	0	mean width ratio 1.50x
crypto	higher recent realised vol raises the predicted median vol	192	0	mean rise in median log-RV for +1 log-RV over 3 days: 0.443

C2. Calibration, Jul 2025 – Nov 2026



x = quantile level, y = share of outcomes below that quantile; on the diagonal = calibrated.

Model	n	max |share − level|	80% coverage
oil/wm_sv	337	0.038	0.766
oil/wm_sv_direct	337	0.031	0.766
oil/rw_gauss	337	0.034	0.801
oil/chronos2	337	0.077	0.825
oil/tirex	337	0.037	0.772
power/wm_power	11,831	0.063	0.758
power/wm_power_blend	11,831	0.017	0.784
power/wm_power_direct	11,831	0.070	0.750
power/lear	11,831	0.048	0.750
power/chronos2	11,831	0.050	0.738
power/tirex	11,831	0.051	0.767
gas/wm_gas_ens	70	0.086	0.843
gas/wm_gas	70	0.143	0.857
gas/wm_gas_direct	70	0.129	0.871
gas/st_ols_wx	70	0.114	0.843
gas/chronos2	70	0.143	0.757
crypto/wm_fin	11,736	0.040	0.840
crypto/wm_fin_blend	11,616	0.053	0.870
crypto/wm_direct	11,736	0.050	0.842
crypto/har_intra_dow	11,616	0.081	0.889
crypto/chronos2	11,760	0.031	0.795

Model	n	max |share − level|	80% coverage
crypto/tirex	11,760	0.038	0.811

All models are compared on the nine shared levels 0.1–0.9. gas: best calibrated wm_gas_ens (max deviation 0.086); wm_gas_ens max deviation 0.086, 80% coverage 0.843; power: best calibrated wm_power_blend (max deviation 0.017); wm_power max deviation 0.063, 80% coverage 0.758; crypto: best calibrated chronos2 (max deviation 0.031); wm_fin max deviation 0.040, 80% coverage 0.840; oil: best calibrated wm_sv_direct (max deviation 0.031); wm_sv max deviation 0.038, 80% coverage 0.766. Coverage below 0.80 means intervals that are too narrow (over-confident); above 0.80, too wide.

C3. Placebo: the true outcomes vs outcomes shuffled across dates (500 shuffles)

If a forecast carried no information about the specific date, shuffling which date each outcome belongs to would not hurt its score. **This is a sanity floor, not a ranking:** every model passes it, including the plain random walk, because shuffling destroys the level/season match. It shows that the scoring is date-specific and that no model's score is an artefact of the pairing. p = 0.0020 is the smallest value 500 shuffles can give (no shuffle scored as well as the truth).

Model	n	QS true	QS shuffled	improvement	p
oil/wm_sv	337	0.009165	0.02405	61.9%	0.0020
oil/wm_sv_direct	337	0.009208	0.02425	62.0%	0.0020
oil/rw_gauss	337	0.0092	0.0241	61.8%	0.0020
oil/chronos2	337	0.009523	0.0237	59.8%	0.0020
oil/tirex	337	0.009435	0.02407	60.8%	0.0020
power/wm_power	11,831	8.601	24.32	64.6%	0.0020
power/wm_power_blend	11,831	8.171	23.81	65.7%	0.0020
power/wm_power_direct	11,831	8.63	24.27	64.4%	0.0020
power/lear	11,831	8.689	23.73	63.4%	0.0020
power/chronos2	11,831	8.151	24.17	66.3%	0.0020
power/tirex	11,831	8.363	23.73	64.8%	0.0020
gas/wm_gas_ens	70	4.277	42.88	90.0%	0.0020
gas/wm_gas	70	4.545	42.85	89.4%	0.0020
gas/wm_gas_direct	70	4.65	42.44	89.0%	0.0020
gas/st_ols_wx	70	6.114	42.02	85.4%	0.0020
gas/chronos2	70	10.15	39.47	74.3%	0.0020
crypto/wm_fin	11,736	0.1491	0.4007	62.8%	0.0020
crypto/wm_fin_blend	11,616	0.148	0.3926	62.3%	0.0020
crypto/wm_direct	11,736	0.1483	0.4005	63.0%	0.0020
crypto/har_intra_dow	11,616	0.1568	0.386	59.4%	0.0020
crypto/chronos2	11,760	0.1593	0.397	59.9%	0.0020
crypto/tirex	11,760	0.1612	0.3935	59.0%	0.0020

Part D — What is still open

- **Sealed ex-ante forecasts:** 126 model-forecasts are sealed (sha256, committed before the outcome); 0 graded so far. The first gas release (2026-11-05 15:30 UTC) and the 2026-11-06 power auction are graded automatically (tools/deploy/score_exante.py, runs/deploy_daily/EXANTE_LEDGER.md).
- **Fresh evidence needs time:** about 20 gas weeks or 60 power/crypto/oil days of forward data before a forward confidence interval is informative. The daily routine (09:13 UTC) accumulates it and audits every run.
- **Re-look caveat:** Part B's 16 months include VAL and TEST, both scored before; only FORWARD (Nov 2026 on) is untouched by any decision.
- **Not tested here:** retraining, new architectures, the HH-price task (no weather data in this environment), and the RL trading agents of Alma-Fin.

How to reproduce

Step	Command
V1 mutation tests	.venv/bin/python tools/validate/mutation_test.py
V2 tamper + determinism	.venv/bin/python tools/validate/tamper_determinism.py
V3 exhaustive leak	.venv/bin/python tools/validate/leak_exhaustive.py
V4 16-month evaluation	ALMA_WINDOWS=posttrain ALMA_DEPLOY_OUT=runs/validation/posttrain .venv/bin/python tools/deploy/deploy_{oil,power,gas,fin}.py
V5 calibration / properties / placebo	.venv/bin/python tools/validate/calib_props_placebo.py
This PDF	.venv/bin/python tools/validate/make_validation_pdf.py